Independencia de variables aleatorias:

$$P_{Y|X}(\beta|\alpha) = P_Y(\beta) \qquad \forall \alpha \in R_X \text{ tal que } P_X(\alpha) > 0$$

$$P_{X|Y}(\alpha|\beta) = P_X(\alpha) \qquad \forall \beta \in R_Y \text{ tal que } P_Y(\beta) > 0$$

$$P_{X,Y}(\alpha,\beta) = P_{Y|X}(\beta|\alpha) \cdot P_X(\alpha) \overset{\text{indep}}{=} P_Y(\beta) \cdot P_X(\alpha)$$

Def.: 2 v.a.d. X, Y son independientes sii

$$P_{X,Y}(\alpha,\beta) = P_X(\alpha) \cdot P_Y(\beta) \qquad \alpha \in R_X, \beta \in R_Y$$

Volvemos al ejemplo de la urna:

$$\left.\begin{array}{l} P_{X,Y}(0,0) = 0.1 \\ P_X(0)\cdot P_Y(0) = 0.5\cdot 0.4 = 0.2 \end{array}\right\} \neq \Rightarrow X,Y \text{ \underline{no} son independientes}$$

Supongamos que X, Y son independientes:

$$E\Big[\overbrace{g_1(X)\cdot g_2(Y)}^{=\,g(X,Y)}\Big] = \sum_{\alpha\in R_X}\sum_{\beta\in R_Y} g_1(\alpha)\,g_2(\beta)\;P_{X,Y}(\alpha,\beta) \overset{\text{indep.}}{=}$$

$$= \sum_{\alpha\in R_X}\sum_{\beta\in R_Y} g_1(\alpha)\cdot g_2(\beta)\cdot P_X(\alpha)\cdot P_Y(\beta) = \overbrace{\Big[\sum_{\alpha\in R_X} g_1(\alpha)\cdot P_X(\alpha)\Big]}^{E[g_1(\alpha)]}\;\overbrace{\Big[\sum_{\beta\in R_Y} g_2(\beta)\cdot P_Y(\beta)\Big]}^{E[g_2(\beta)]}$$

X, Y son indep.

$$\boxed{E\big[g_1(X)\cdot g_2(Y)\big] = E\big[g_1(X)\big]\cdot E\big[g_2(Y)\big]}$$

X, Y son indep.: $E[X \cdot Y] = E[X] \cdot E[Y] = \mu_X \cdot \mu_Y$

$$Cov(X,Y) = E[X \cdot Y] - \mu_X \cdot \mu_Y = \mu_X \mu_Y - \mu_X \cdot \mu_Y = 0$$

$$\boxed{\begin{array}{c} X, Y \text{ son indep.} \Longrightarrow Cov(X,Y) = 0 \\ \rho_{X,Y} = 0 \end{array}}$$

¡Ojo!: no vale al revés: $Cov(X,Y) = 0 \not\Rightarrow X, Y$ indep.